

1. Executive Summary

The objective was to diagnose frequent operational disruptions. By positioning a diagnostic machine between the target user and the gateway using **ARP poisoning**, I captured live traffic to analyze network health. The analysis revealed significant **packet loss** and **retransmission errors** during secure web interactions, identifying physical or congestion-layer instability as the root cause.

2. Technical Walkthrough

Phase 1: Intercepting Network Traffic (ARP Spoofing)

To analyze the traffic, the diagnostic machine was positioned as a "Man-in-the-Middle" to intercept packets between the target workstation (192.168.18.73) and the network gateway (192.168.18.1).

- **Gateway Poisoning:** I used arpspoof to tell the target machine that my Kali Linux machine was the gateway.
- **Target Poisoning:** I simultaneously told the gateway that my machine was the target workstation.
- **Result:** This bidirectional spoofing ensured all business traffic passed through the sniffer for analysis.

[illegible]

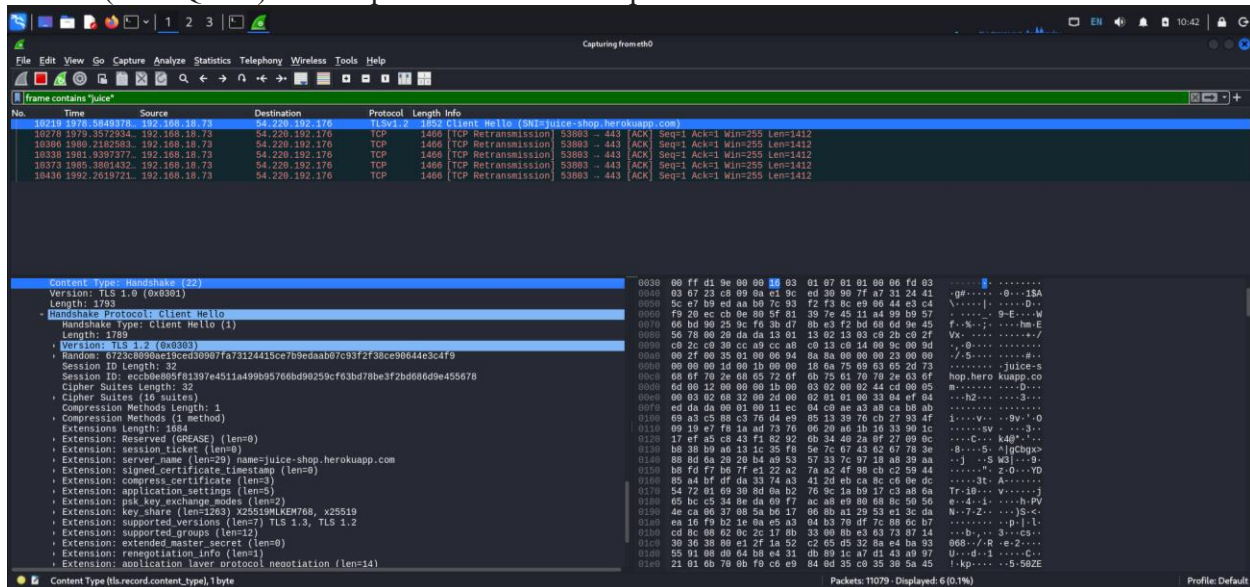


- **Filter Used:** frame contains "juice".
- **Observation:** I identified a TLS 1.2 "Client Hello" handshake from the target to juice-shop.herokuapp.com. This confirmed the specific session being disrupted.



The sniffing process revealed critical errors in the transport layer:

- **Packet Loss:** Wireshark flagged multiple instances where a "**Previous segment not captured**". This indicates the network dropped data packets before they reached the destination.
- **High Latency & Retransmissions:** The capture shows a series of "**TCP Retransmission**" errors. The workstation was forced to resend the same data repeatedly because the network failed to deliver it or acknowledge receipt.
- **Root Cause:** These symptoms point to **congested network hardware** or **signal interference** (if using Wi-Fi), which prevents the reliable delivery of the encrypted (TLS/QUIC) data required for business operations.



4. Proposed Solutions

To resolve the disruptions and stabilize operations, I recommend the following:

1. **Deploy Wired Connections:** Transition critical workstations from Wi-Fi to **shielded Ethernet cables** to eliminate the interference causing the "segment not captured" errors.
2. **Hardware Upgrade:** The frequent retransmissions suggest the current router or switch is struggling with the processing load of modern encrypted traffic. Upgrading to a **high-throughput enterprise switch** is advised.
3. **Implement QoS:** Configure **Quality of Service (QoS)** on the router to prioritize business-critical web traffic (TLS/HTTPS) over non-essential background data.